Feasibility Report

Disaster Response Resource Allocation System

Software Engineering — 3rd Semester Project

Project Title	Disaster Response Resource Allocation System
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1. Technical Feasibility

Technical feasibility assesses whether the **Disaster Response Resource Allocation** system can be built with the technologies currently accessible to the team and suited to the anticipated workload.

Key technical components of the proposed system include multiple database queries, map-based area virtualization, and optimization techniques for allocating resources based on factors such as severity, distance, population and the availability of resources. The system can be developed using commonly available web technologies for the frontend, backend, database, and map integration, making the required development environment accessible to the team.

The database will be used to store information related to affected areas, available resources, location, population and allocation result. The map based interface will be used for visualization of affected areas and possible routes between them.

The optimization component will process the available information and generate suitable resource allocation decisions while considering the defined factors.

The proposed system is primarily intended as a situation and decision support application. So it does not require specialized hardware for initial implementation. The workload of the system can be managed on a standard hardware and does not require a specialized hardware or system, it can be tested on a public dataset or a simulated dataset.

Overall the technology, development environment and computational resources are available . So this project is completely feasible.

Technologies and tools

- **Backend:** Node.js with Express, or Python with FastAPI, for exposing allocation and scenario-management APIs.
- **Optimization engine**: an open-source solver such as Google OR-Tools or PuLP (Python) for the optimization-based allocation, compared against a simple greedy/nearest-available baseline implemented natively.
- **Distance/Travel-time estimation:** OpenStreetMap data accessed through the free OSRM (Open Source Routing Machine) API, avoiding dependency on paid mapping services.
- **Database:** PostgreSQL (with PostGIS extension for spatial queries) or MongoDB, both free and open-source.
- **Frontend/Dashboard:** React with a mapping library such as Leaflet.js for visualizing areas, resource bases, and allocation results on a map.

2. Operational Feasibility

Operational feasibility checks whether the system can actually be used easily by the people it is made for and whether it fits into how they work.

The system is meant to help emergency planners or coordinators decide how to send limited resources like ambulances and rescue teams to different affected areas. It does not replace their decisions, it only gives them a suggestion and shows it clearly on a map. A user only has to enter or load the scenario details, run the allocation, and see the result on the dashboard. The dashboard shows the affected areas, resource bases, and the allocation on a map, along with simple numbers like response time and unmet demand, so a person without technical knowledge can understand it.

The comparison between the simple nearest-resource method and the optimized method also helps users see why the suggested allocation is different. This makes the system easier to trust.

The system does not need any complicated setup for the users. It runs in a web browser, so nothing special has to be installed on their side. The team can handle the maintenance for a prototype like this, such as updating sample data and fixing bugs. Since real emergency data is not available for a college project, simulated data will be used and it will be clearly mentioned. So the system is operationally feasible as a decision support prototype.

3. Economic Feasibility

Economic feasibility checks whether the cost of building the project is reasonable and affordable for the team.

Almost all the tools in this project are free and open source. Python, FastAPI, OR-Tools or PuLP, PostgreSQL with PostGIS, React, Leaflet.js, OpenStreetMap and OSRM can all be used without paying anything. Because of this, there is no need to buy any software licences or paid map services.

No special hardware is needed either. The system can be developed and tested on the normal laptops the team already has. Public datasets and simulated data will be used, so there is also no cost for data.

The main cost is the time and effort of the team members. If the project is ever hosted online for a demo, it can be done on a free or low-cost hosting plan, which is a very small amount.

Estimated costs:

- Software and libraries: free (open source)
- Map and routing data: free (OpenStreetMap and OSRM)
- Hardware: no extra cost, existing laptops are enough
- Data: free (public and simulated datasets)
- Hosting (optional, for demo only): free tier or a very small cost